properties like strength, permeability and refractoriness. The following three types of additives are used.

Reducing Agents: This type of additive prevent the formation of oxide. They fill up the fine pores in the mould surface. This improves the surface finish on the casting. Coal dust, fuel oil and sea coal are some reducing agents.

Fibrous Material: This type of additive prevent the formation of dry surface on the mould. It improves the collapsibility of mould. Wood flour, straw, cow dung, asbestos and saw dust are some fibrous materials.

Special Additive: Some special additive are added to improve the dry strength and collapsibility of mould. These additive also prevent quick drying of the mould. Dextrin and molasses are some special additives.

6.9 SAND INGREDIENTS - CLASSIFICATION OF MOULDING SAND

Moulding sand are classified as follows:

1. Natural Sand
2. Synthetic Sand
3. Special Sand

Natural Sand or Green Sand: Natural sand is available at river beds. It contains 80 to 90% silica, 5 to 10% alumina or clay and small amount of lime and magnesia.



Green Sand

Natural sand is used to make casting in ferrous and non-ferrous metals.

Synthetic Sand: Synthetic sand is prepared to obtain required properties by adding some ingredients with the natural



Synthetic Sand

sand. Bentonite, water, iron, oxide, calcium and magnesium are mixed with natural sand. Synthetic sand is used in machine moudling and high pressure moulding.

Special Sand: Special sands are prepared to obtain specific properties such as



Special Sand

refractoriness, high heat conductivity and low expansion. Good quality casting with fine surface finish can be produced by using special sands. The most widely used special sands are given below.

Oilvine Sand: It is in green color. It has medium refractoriness. It is used for producing non-ferrous castings of complicated shapes.

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